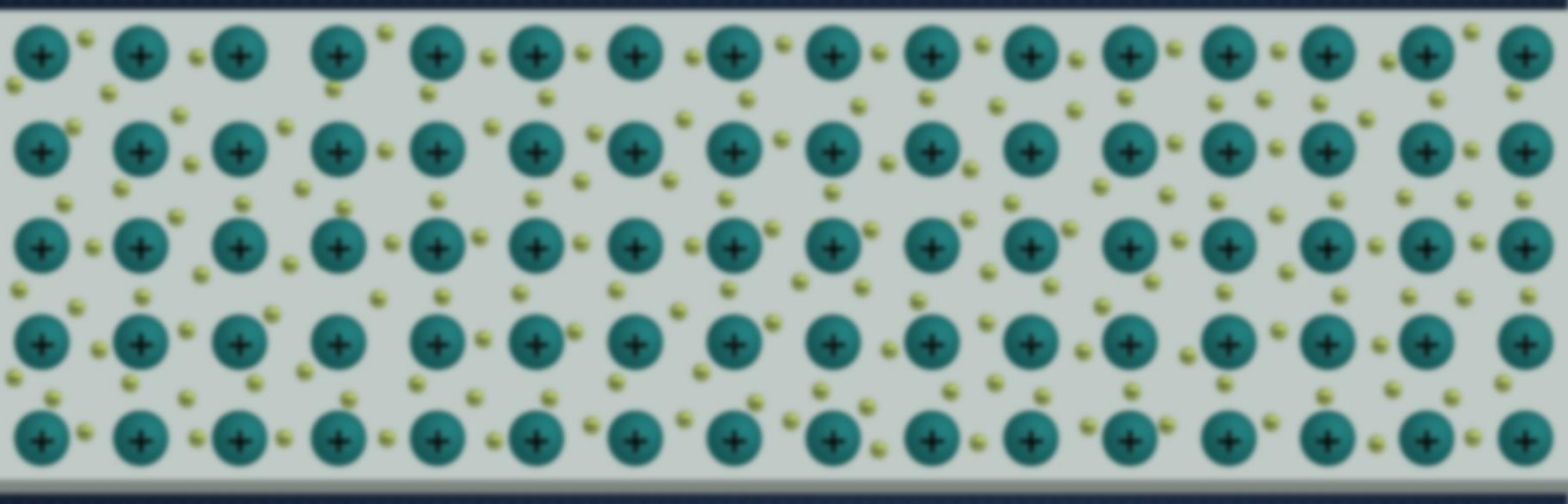

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Physical Operations in Superconducting Qubits

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September 2024

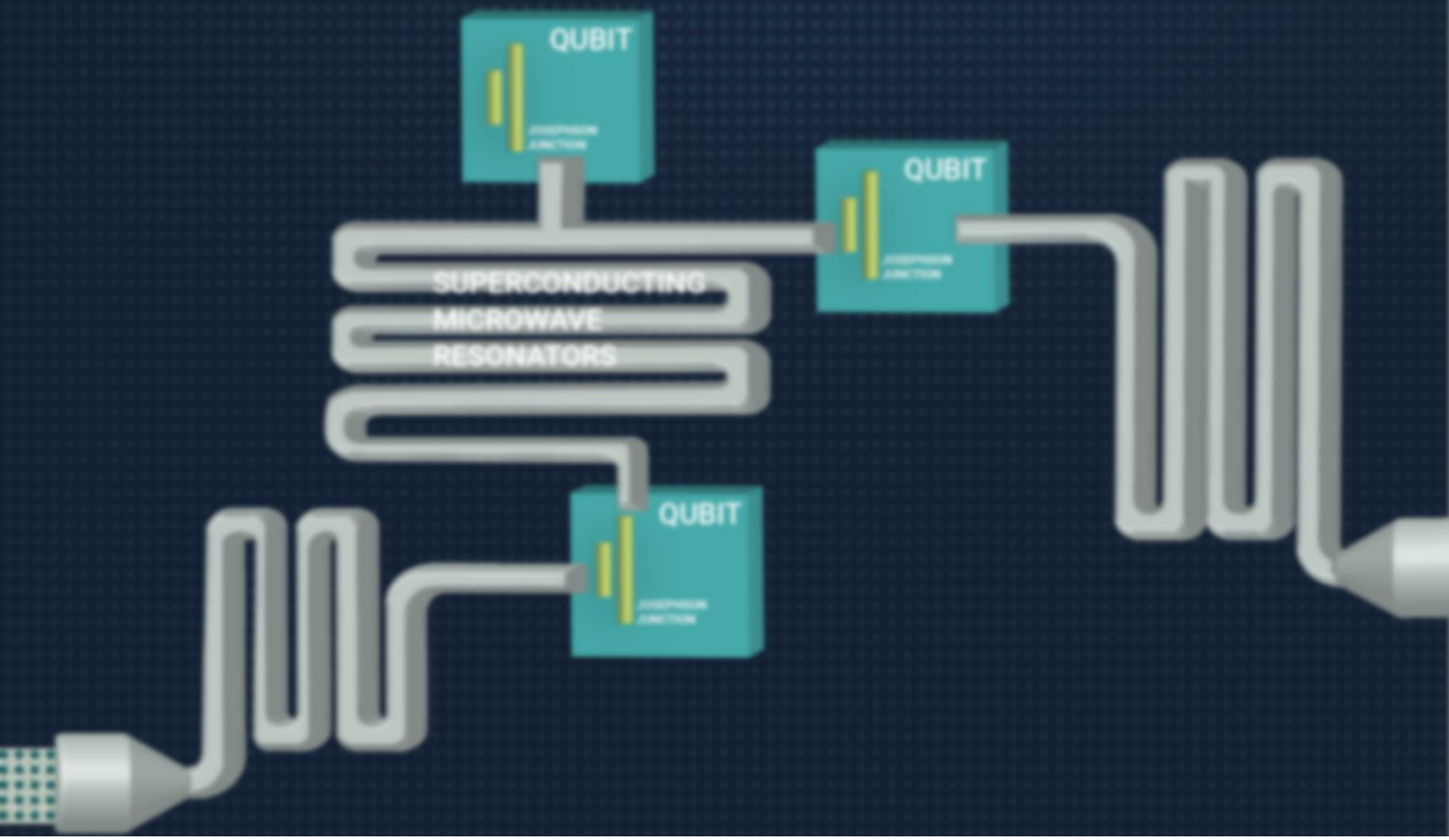

TABLE OF CONTENTS

Introduction 4
Solid State Basics 5
Bloch's Theorem 5
Conductivity: isolators, semiconductors, and metals 5
Optical Properties of Materials 7
Electrical Resistance and Temperature 7
Superconductivity 9
Properties of Superconductivity 9
Cooper Pairs and Condensate Wavefunction 10
Types of Superconducting Qubits 11
Josephson Junction 11
Transmon Qubits (Charge-based) 14
Flux Qubits 19
DiVicenzo Criteria 24
Initialising qubits to a simple fiducial state 24
Long relevant Quantum coherence Times 24
Universal set of Quantum Gates 24
Qubit specific measurement capability 24
Interconverting stationary and flying qubits and faithfully transmitting flying qubits between specified locations 24
Scalable Physical System with well-characterized qubits 25
Key Concepts 26
Bibliography 28

INTRODUCTION

This article provides an overview of how the electrical circuits that constitute superconducting qubits operate. To explore this topic, I will address several key questions, including:

- What are the most common types of superconducting qubits? How do they operate?
- How does the platform implement the states $|0\rangle$ and $|1\rangle$ for a qubit? What physical phenomenon are they using?
- How is the state $|0\rangle$ initialized? What is the typical error? How are operations on a single qubit implemented?

By examining these aspects, the goal is to offer a comprehensive understanding of the underlying mechanisms and typical challenges associated with superconducting qubits.

It might be important to mention that this article is organised to be accessible to any Bachelor's-level student. My goal is to present these concepts in a straightforward way, breaking down explanations when necessary, so that anyone interested in quantum circuits can better understand them, regardless of their prior knowledge. Therefore, I will start with some solid-state basics before moving on to superconductors, a topic typically covered at the Master's level.

To keep the main text concise, some important concepts that are not fully explained here are covered in the 'Key Concepts' section at the end of the paper. These words will be underlined in green: feel free to refer to that section for more details and definitions as needed.

SOLID STATE BASICS

Do you have a *solid* understanding of Condensed Matter Theory? Great! You can skip this chapter. Here, we will cover some basics to better understand how electrons behave in solids, as well as their magnetic and optical properties. This background is important for gaining an overview of how superconductors function since it lays the foundation for understanding what goes on in a qubit electrical circuit.

Bloch's Theorem

Since the nearly free electron model will be used to describe materials in terms of conductivity, it is best to explain Bloch's Theorem. That being said, what is meant by "*nearly free*" electron model?

Let us start by imagining that the electrons in a crystal are weakly perturbed by a periodic potential. In real materials, the scattering from atoms can be extraordinarily strong, so much so that perturbation theory will prove inaccurate to describe electrons. To describe them as plane waves, we must understand that their wave momentum is not conserved: however, regardless of the periodic potential's strength, the crystal's momentum might be, in fact, conserved.

As long as the potential is periodic (in the unit cell), crystal momentum, $\vec{k}$, is conserved. This means that we are able to describe electron eigenstates in terms of the crystal momentum they correspond to.

$$\psi_k(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, u_{\boldsymbol{k}}(\boldsymbol{r}) \quad (1)$$

Bloch's Theorem, where Ψ is the modified plane wave, ***r*** *is the electron's position vector,* ***k*** *is the crystal momentum vector and u is a periodic function (sometimes known as Bloch's function).*

Now that we understand why it is possible to represent electrons using crystal momentum, we may move on to conductivity.

Conductivity: isolators, semiconductors, and metals

A material's conductivity can be determined by looking at their energy band graphs. These graphs show which states, in a certain $\boldsymbol{k}$ range, are completely or partially occupied by electrons and the states' respective energy.

The **Fermi Energy,** $\boldsymbol{E_F}$, corresponds to the energy of the highest occupied state at absolute zero, which means that electrons will occupy all the states, from lower to highest, until they reach E_F. Since below this value all states are full, it is sufficient to represent only their valence band. Let us take a look at the different possibilities of energy bands.

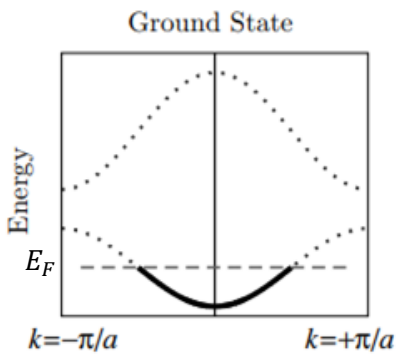

Figure 1 – Partially filled energy band, characteristic of metals.

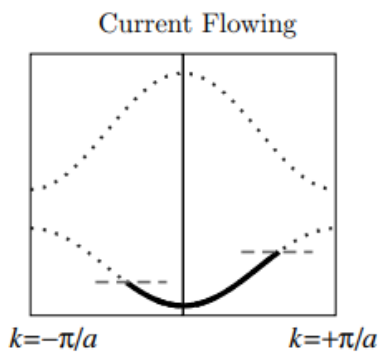

Figure 2 - Shift caused by an electrical field.

When an electric field is applied to a partially filled energy band, the electrons tend to shift from occupied energy states to empty, more energetic, ones, allowing current to flow. This type of materials are **metals** and, at room temperature, they're typically the most conductive. The effect of applying an electrical field in a metal can be seen on the right.

Even with enough electrons to completely fill an energy band, two situations might arise:

- All energy states in the lower energy band are occupied
- The energy of two different bands overlap

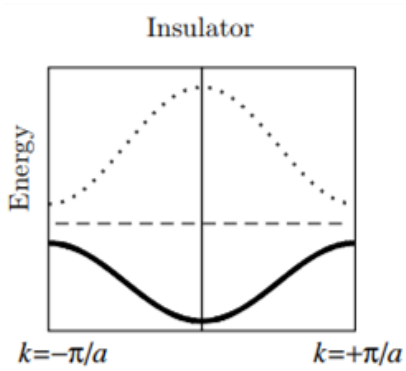

Figure 3 – Filled energy band, characteristic of isolators.

In the first case, the lower, entirely filled band is known as the **valence band**, while the upper empty band is known as the **conduction band**. Moving an electron from the valence to the conduction band creates the minimum energy excitation, which is non-zero. At absolute zero, these systems do *not* respond to electrical fields, which is why materials with this property are called **insulators**, band insulators to be specific. However, if the band gap is below 4 eV, a fraction of the material's electrons can be thermally excited at room temperature, transitioning to the conduction band. For this reason, these later materials are called **semiconductors**, which are a type of insulator.

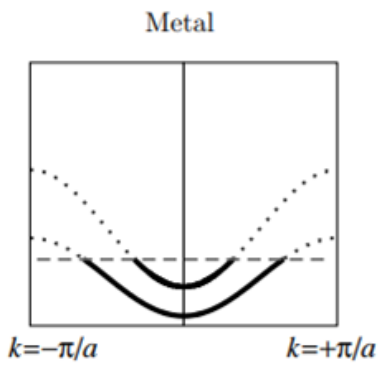

Figure 4 – Electrons distributed between two overlapping energy bands.

If, however, the band energies overlap, the electrons will be distributed by two bands' states bellow the Fermi Energy. This means that both bands contain empty states and, like we've seen earlier, partially complete bands allow current to flow. This means that this kind of overlap generates a metal.

It is worth noting that since electrons are fermions, they are distinguishable particles. The Pauli Exclusion Principle states that fermions can't occupy the same quantum state, causing electrons to filling available states from lower to higher energy until reaching Fermi Energy.

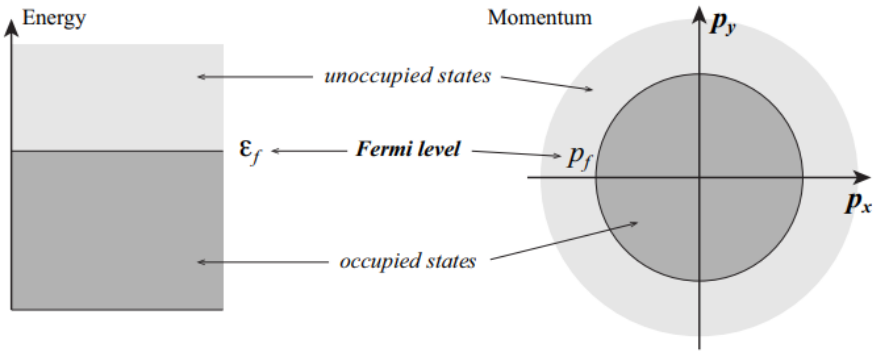

Figure 5 – Electronic band structure in a metal.

In conclusion, the number of valence electrons and the structure of the bands themselves are key factors to determining a materials conductivity. Let us take a look at optical properties.

Optical Properties of Materials

Understanding the interaction between different material types and photons is crucial for comprehending how energy is exchanged in a system during electrical operations.

Since band insulators have a full valence band, a photon must possess energy equal to the gap between the valence and conduction bands to be absorbed. Otherwise, its energy will be insufficient to excite an electron to the conduction band, allowing the photon to pass through the material unchanged.

Semiconductors with smaller band gaps are able to absorb photons with energies above the band gap, able to excite electrons into the conduction band. Photons with an energy bellow the band gap won't be absorbed.

Since metals are very conductive, electrons are easily excited by photons, re-emitting light when returning to their ground state.

Electrical Resistance and Temperature

Electrical resistance in conductors depends on temperature, but this relationship varies based on the type of material. For insulators, resistance is intuitively very high due to the tight bond between electrons and their atoms. However, semiconductors and metals exhibit distinct temperature-dependent behaviours that merit closer examination.

For metals, the relationship between the two quantities can be **linearly approximated**, with the resulting equation being valid within a range that excludes temperatures near absolute zero or close to the metal's melting point.

$$R(T) = R_0[1 + \alpha(T - T_0)] \quad (2)$$

Linear approximation for the temperature dependence of resistance. R is the conductor resistance at temperature T, in Celsius, R_0 is the conductor resistance at a reference temperature, α is the temperature coefficient of resistance, and T_0

Keep in mind that resistance and resistivity are not the same: resistivity, ρ, is an intrinsic property of the material, while resistance depends on both the material's resistivity, the length, L, and cross-sectional area of the object, A.

$$R = \rho \frac{L}{A} \quad (3)$$

Relation between resistance and resistivity.

From this equation we get that **resistivity in metals rises in higher temperatures**. This is due to the fact that the number of phonons increases with temperature, which means increased electron-phonon scattering, resulting in energy loss. As electrons collide with phonons, they transfer kinetic energy to the lattice, which slows down their movement and, therefore, results in increased resistivity

While resistivity in metals is relatively simple, it becomes more complex in semiconductors, since charge carriers can differ in nature. An approximation is then more appropriate for explaining how temperature affects resistivity.

$$\rho(T) \propto e^{\left(\frac{E_g}{2k_B T}\right)} \quad (4)$$

Temperature dependence on resistivity. E_g represents the energy gap between the valence and conduction bands, k_B is the Bolzmann constant and T is the temperature in Kelvin.

Resistivity in semiconductors decreases exponentially as temperature increases due to the greater number of thermally excited charge carriers. Higher temperatures provide more energy to excite electrons from the valence band to the conduction band, reducing resistivity. It is important to note that the rate at which resistivity decreases with temperature depends on the semiconductor's band gap energy; larger band gaps result in slower decreases in resistivity with rising temperature.

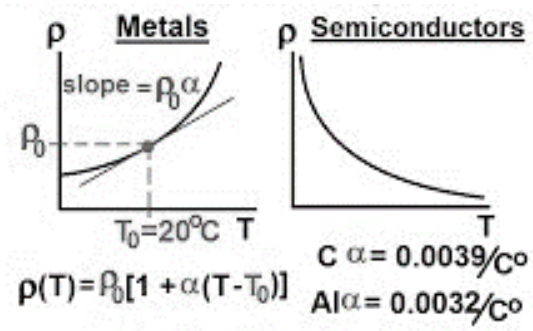

Figure 6 – Temperature dependence of Resistivity in metals and semiconductors.

In conclusion, the way temperature affects resistivity is different in metals and semiconductors. For metals, resistivity increases linearly with temperature, meaning high resistivity at higher temperatures. In contrast, semiconductors' resistivity lowers as temperature rises exponentially. This contrast highlights how temperature influences the electrical properties of these materials differently: increasing resistivity in metals and decreasing it in semiconductors.

SUPERCONDUCTIVITY

Like the previous chapter, this one is meant as an overview of a few theoretical concepts about superconductivity. If you are already familiar with this topic, feel free to skip as well.

Properties of Superconductivity

Bardeen–Cooper–Schrieffer (BCS) Theory provides a microscopic explanation for superconductivity in certain materials at low temperatures. In order to get an overview of how superconductors differ from regular conductive materials, we are going to extract some key properties from it.

Critical Temperature and Electrical Resistance

Superconductivity refers to the phenomenon in which a material exhibits **zero electrical resistance**. It only occurs below a certain **critical temperature,** $\boldsymbol{T_c}$, which depends on the type of material. Even though this temperature typically sits close to absolute zero, there have been discoveries of materials that show superconductivity at high temperatures, that is, above 77 K.

Zero electrical resistance means that electron scattering is virtually eliminated, as electrons form **Cooper pairs**, which will be explained further below. Lattice vibrations, or phonons, no longer cause significant interactions with these Cooper pairs, allowing electrons to flow smoothly through the material, avoiding atom collision as well.

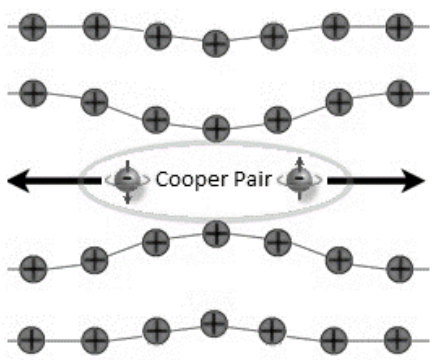

Figure 7 – Cooper pair in a lattice.

Cooper pairing significantly prevents electron scattering and atomic collisions, which are the usual mechanisms of resistance. This leads to superconductors perfect electrical conduction, preventing energy loss.

Meissner Effect

When a material transitions into a superconducting state, it begins to expel all magnetic fields from its interior: this phenomenon is known as Meissner effect. When in the presence of a magnetic field, the superconductor will develop currents on its surface that generate a counter magnetic field which cancels out the applied field inside the material.

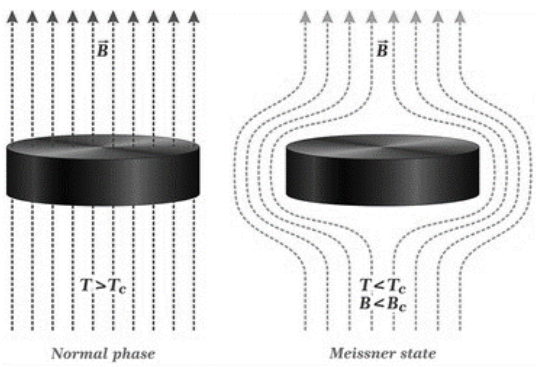

Figure 8 – Meissner Effect.

Superconductors aren't, however, perfect conductors. Repelling all magnetic fields bellow their critical temperature means that as the magnetic field lines are pulled out of the material, the interior is left completely free of any magnetic influence.

Another fundamental concept in superconductivity is **critical magnetic field**, B_c, which represents the threshold beyond which a superconductor can no longer maintain its superconducting state. When exposed to

magnetic fields weaker than this limit, the material expels the field through the Meissner Effect, maintaining zero resistance. However, once the magnetic field strength reaches the critical point, the superconductor reverts to its normal, resistive state, losing its unique properties. This limit varies depending on the material and its temperature.

Cooper Pairs and Condensate Wavefunction

An oversimplified explanation of the attraction between two electrons can be understood by considering what happens as one electron moves through the lattice. As it passes by the positively charged atomic nuclei, the electron causes a slight displacement of the nuclei in its direction, creating a disturbance in the lattice. This localized buildup of positive charge attracts a second passing electron, **weakly binding** it to the first electron. This binding is extremely delicate, which is why it only occurs at cryogenic temperatures: lower temperatures mean fewer phonons, reducing interactions with the electrons.

The coupling between the opposite-spin electrons in a Cooper Pair results in a total spin of zero, since one electron has spin $s_1 = +{}^{1}/_{2}$ and the other one $s_2 = -{}^{1}/_{2}$. Total momentum is also zero, since their momenta is opposite as well. Although individual electrons are fermions, when they form a Cooper Pair, they start behaving as **bosons**, adopting new properties.

Pauli's Exclusion Principle does not apply to bosons. For this reason, the Cooper Pairs in the superconductor form a giant condensate which represents **a collective ground state** for the system. In the superconducting condensate, the wavefunction of each Cooper Pair can be written as:

$$\Phi(\boldsymbol{r_1}, \boldsymbol{r_2}) = \sum a_j \, \phi(\boldsymbol{r_1}, \boldsymbol{P}_j, \uparrow)\phi(\boldsymbol{r_2}, -\boldsymbol{P}_j, \downarrow) \quad (5)$$

Cooper Pair wavefunction, where $\phi(\boldsymbol{r_1}, \boldsymbol{P}_j, \uparrow)$ is the single electron wave function with momentum $\mathbf{P}$ and spin up.

Similar to metals, electrons in superconductors still face scattering phenomena. The difference is that the Cooper-pair wavefunction has **strong symmetry**, ensuring that the scattering events are symmetrical as well. For there to be an excitation in the superconducting ground state, the Cooper Pair must be broken, resulting in two singular unpaired electrons that break the symmetry in the system. To do this, it is necessary to overcome an energy barrier, creating an energy gap between ground states and excited states.

This wavefunction represents a superposition of states with many different momenta. Since total momentum is conserved during scattering events, the wavefunctions suffers little changes. Since it already counts for the superposition of every state, the overall quantum state of the pair remains unaffected, meaning **no energy losses** from the system.

TYPES OF SUPERCONDUCTING QUBITS

Although most superconducting qubits use a component called Josephson Junction, the different elements in the circuits are what determines how qubits' quantum states are defined. For this reason, we must understand the behaviour of the different types of qubits before learning how operations are implemented.

The generalised coordinates used to write the Hamiltonian of each system are flux, ϕ, and charge, c, for both qubit types.

Josephson Junction

Superconducting qubits have a characteristic behaviour of a nonlinear harmonic oscillator. This behaviour can be attributed to the Josephson Junction.

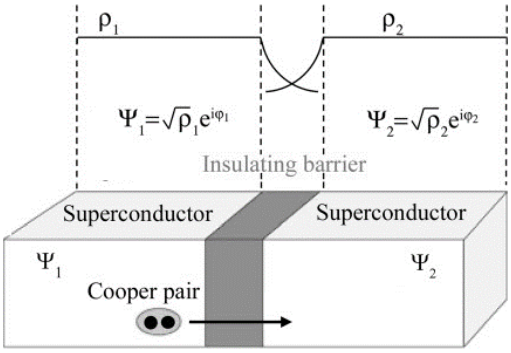

Figure 9 – Schematic of a Josephson Junction.

Junction Wavefunction

This component consists of two superconductors, S_1 and S_2, separated by an insulating barrier, I. The equations of the Cooper Pair present in each superconductor, respectively, can be written as:

$$\Psi_1(\boldsymbol{r}) = \sqrt{\rho_1}\, e^{i\phi_1(\boldsymbol{r})}, \text{ in } S_1 \quad (6)$$

$$\Psi_2(\boldsymbol{r}) = \sqrt{\rho_2}\, e^{i\phi_2(\boldsymbol{r})}, \text{ in } S_2 \quad (7)$$

Ψ_1 and Ψ_2 are the macroscopic wavefunctions of superconductors S_1 and S_2, respectively, ρ_1 and ρ_2 are the local Cooper pair densities in S_1 and S_2 , ϕ_1 and ϕ_2 are the phase of each superconductor's wavefunction, and $\boldsymbol{r}$ is the spatial coordinate.

The phase difference between the two superconductors is what defines the junction's behaviour, since it dictates the supercurrent that can tunnel through the isolator. The junction's wavefunction is a coherent superposition of the states in S_1 and S_2.

$$\Psi_I(\boldsymbol{r}) = \alpha\, \Psi_1(\boldsymbol{r}) + \beta\, \Psi_2(\boldsymbol{r}) \quad (8)$$

We can recall equations (6) and (7) to further develop the junction's wavefunction.

$$\Psi_I = \alpha \sqrt{\rho_1}\, e^{i\phi_1(\boldsymbol{r})} + \beta \sqrt{\rho_2}\, e^{i\phi_2(\boldsymbol{r})}$$

For simplicity, let's assume that the density of Cooper pairs is the same in both superconductors, so that $\rho_1 = \rho_2 = \rho$.

$$\Psi_I = \sqrt{\rho}\, (\, \alpha\, e^{i\phi_1(\boldsymbol{r})} + \beta\, e^{\,i\phi_2(\boldsymbol{r})}\,)$$

Since Ψ_1 and Ψ_2 are part of a coherent superposition, the coefficients α and β must reflect the relative phase between the two wavefunctions. Let us assume that the two superconductors have equal contribution in the final wavefunction. This would mean that $|\alpha| = |\beta|$, and therefore, in order to ensure normalization, $|\alpha|^2 + |\beta|^2 = 1$, which leads us to $|\alpha| = |\beta| = 1/\sqrt{2}$. We now have:

$$\Psi_I = \frac{\sqrt{\rho}}{\sqrt{2}} \left(e^{i\phi_1(\boldsymbol{r})} + e^{i\phi_2(\boldsymbol{r})} \right)$$

Given phase difference being $\Delta\phi(\boldsymbol{r}) = \phi_1(\boldsymbol{r}) - \phi_2(\boldsymbol{r})$, we can use this to factor one of the exponentials. Let's factor ϕ_2 out.

$$\Psi_I = \frac{\sqrt{\rho}}{\sqrt{2}} e^{i\phi_2(\boldsymbol{r})} \left(\frac{e^{i\phi_1(\boldsymbol{r})}}{e_2^{i\phi(\boldsymbol{r})}} + 1 \right) =$$

$$= \frac{\sqrt{\rho}}{\sqrt{2}} e^{i\phi_2(\boldsymbol{r})} \left(e^{i(\phi_1(\boldsymbol{r}) - \phi_2(\boldsymbol{r}))} + 1 \right) =$$

$$= \frac{\sqrt{\rho}}{\sqrt{2}} e^{i\phi_2(\boldsymbol{r})} \left(e^{i\Delta\phi(\boldsymbol{r})} + 1 \right)$$

The resulting wavefunction is, therefore, given by:

$$\Psi_I = \frac{\sqrt{\rho}}{\sqrt{2}} e^{i\phi_2(\boldsymbol{r})} \left(e^{i\Delta\phi(\boldsymbol{r})} + 1 \right) \quad (9)$$

Wavefunction in the junction. The term $\left(e^{i\Delta\phi(\boldsymbol{r})} + 1 \right)$ describes the interference between the two wavefunctions that origins from their phase difference.

Josephson Current

The Cooper pairs that are able to pass through the junction by **quantum tunnelling** can be quantified in what is called **Josephson Current, $\boldsymbol{I}$**. Josephson current is a quantum mechanical **supercurrent**, meaning it flows without resistance through a superconductor due to quantum effects. Electric charge can flow continuously without energy loss as long as the supercurrent remains below a threshold: the **critical current, $\boldsymbol{I_c}$,**which is the maximum supercurrent that can flow through the junction without causing a breakdown of the superconducting state. The phase dependence of supercurrent can be seen in the Josephson Current Equation.

$$I = I_C \sin(\Delta\phi) \quad (10)$$

Josephson Current Equation. When $\Delta\phi = 0$, the two wavefunctions are in phase, and the maximum possible current flows through the junction. When $\Delta\phi = \pi$, they are exactly out of phase, and no current flows through the junction.

Energy in the Junction

The charge, Q, in the junction is related to the phase difference. They have the following relation:

$$Q = C \frac{d\Delta\phi}{dt} \quad (11)$$

Charge in the junction. C being the capacitance of the junction.

Since the current is defined as the time derivative of charge, we can utilize equation (11) to derive its expression.

$$I = \frac{dQ}{dt} = \frac{d}{dt}\left(C\frac{d\Delta\phi}{dt}\right) = C\left(\frac{d^2\Delta\Phi}{dt^2}\right) \quad (12)$$

In general terms, the work done, W, by an electric current as it moves through a potential difference, dV, is given by:

$$W = \int I\, dV \quad (13)$$

In the case of a Josephson junction, instead of a potential difference, dV, we consider the variation in the phase difference across the junction, $d\Delta\phi$. The work done can be interpreted as the energy associated with changing the phase difference in the junction, E_J. As convention, the negative sign indicates that when the current does work against the change in phase difference, energy is stored in the system.

$$E_J = -\int I\, d\Delta\phi \quad (14)$$

Substituting the expression for Josephson Current in (14), we get:

$$E_J = -\int I_C \sin\Delta\phi\ d\Delta\phi \ \rightarrow E_J = -I_C(-\cos\Delta\phi) + E_0$$

Setting $E_0 = 0$, for simplicity, we get to the conventional energy form:

$$E_J = -E_{J0}\cos(\Delta\phi) \quad (15)$$

Energy at the Josephson Junction, where $E_{J0} = I_c$. This form emphasizes that the energy is negative, indicating a stable state at $\Delta\phi = 0$.

Nonlinear Oscillator Model

To move towards a nonlinear oscillator model, we may start by constructing the Lagrangian, $\mathcal{L}$, of the system. The Lagrangian is defined as:

$$\mathcal{L} = T - U \quad (16)$$

T is the kinetic energy related to the motion of the phase and U is the potential energy given by E_J.

Since we defined the effective velocity, in this context, to be the derivative of phase, $\Delta\dot{\phi}$, we can substitute it in the classical kinetic energy expression.

$$T = \frac{1}{2}C\left(\Delta\dot{\phi}\right)^2 \quad (17)$$

Here, the "mass" equivalent in the context of a superconducting circuit is given by the capacitance associated with the charge in the junction, C.

Using the energy in the junction as our potential energy, our system's Lagrangian ends up as:

$$\mathcal{L} = \frac{1}{2}C\left(\Delta\dot{\phi}\right)^2 + E_{J0}\cos(\Delta\phi) \quad (18)$$

A Lagrangian equation of motion has the following general expression:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0 \quad (19)$$

In this equation, q *represents the general coordinate.*

Using the phase difference as the general coordinate, we may now substitute the expression of the Lagrangian into the previous equation. Let's analyse the different terms.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \Delta\dot{\phi}}\right) = \frac{d}{dt}\left(C\,\Delta\dot{\phi}\right) = C\Delta\ddot{\phi}$$

$$\frac{\partial \mathcal{L}}{\partial \Delta\phi} = -E_{J0}\sin(\Delta\phi)$$

Computing this in equation (19) leaves us with a second-order differential equation.

$$C\,\frac{d^2\Delta\phi}{dt^2} + E_{J0}\sin(\Delta\phi) = 0 \quad (20)$$

Euler-Lagrangian equation of the Josephson junction.

For small values of $\Delta\phi$, we can approximate $\sin\Delta\phi \approx \Delta\phi$. Within this approximation, the system behaves as **simple harmonic oscillator**:

$$C\Delta\ddot{\phi} + E_{J0}\Delta\phi = 0\,, \text{with } \omega_0 = \sqrt{\frac{E_{J0}}{C}} \quad (21)$$

A standard harmonic linear oscillator can me modified into a nonlinear oscillator. The simplest form of a nonlinear oscillator is:

$$\ddot{x} + \omega_0^2 x + \alpha x^n = 0 \quad (22)$$

Equation of a nonlinear oscillator, where ω_0 *is the natural frequency, the coefficient* α *quantifies the nonlinear behaviour and* n *is a natural number that defines the degree of nonlinearity.*

With this in mind, it becomes clear that for larger phase differences, the sine function in the Lagrange equation introduces nonlinearity into the system. As a result, **the Josephson junction behaves as a nonlinear oscillator**.

This nonlinear oscillator behaviour is a fundamental characteristic of qubits based on Josephson junctions.

Transmon Qubits (Charge-based)

Transmon Qubits evolved from earlier versions of qubits, called Cooper-pair Box Qubits, or CPBs. Their improved form was designed in a way that reduces charge sensitivity by increasing the capacitance in the circuit. The transmon then spreads the charge over a larger area, making it less affected by small changes in environmental charge.

A transmon qubit is composed of a Josephson junction connected in parallel with a capacitor. Our objective is to derive the Hamiltonian of this system, which can be formulated in terms of the charge and phase variables across the junction.

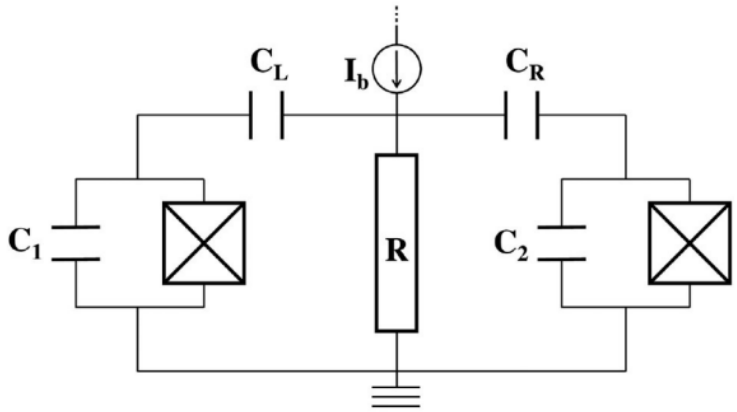

Figure 10 - Circuit diagram representing two transmon qubits. Each qubit is characterized by the capacitances C1 and C2, respectively, capacitively coupled to the same resistor $\boldsymbol{R}$, through the capacitors $\boldsymbol{C_L}$ and $\boldsymbol{C_R}$.

Phase Difference and Magnetic Flux

It is beneficial to describe the phase difference across the Josephson junction in terms of magnetic flux. For this reason, we will use the following relation:

$$\Delta\phi = \frac{2\pi\Phi}{\Phi_0} \quad (23)$$

Josephson flux-phase relation, where $\Delta\phi$ is the phase difference across the Josephson junction, Φ is the magnetic flux through the superconducting loop, $\Phi_0$is the magnetic flux quantum.

Where

$$\Phi_0 = \frac{h}{2e} \quad (24)$$

Flux quantum definition, where h is the Planck constant and e is the elemental charge of the electron. The term $2e$ comes from the fact that superconducting current is carried by Cooper-pairs, each one composed of two electrons.

Equation (22) describes how the **phase difference** across a Josephson junction is **proportional to the magnetic flux** threading the superconducting loop. The phase difference directly controls the supercurrent flowing through the junction, making this relation fundamental to understanding the behaviour of flux-based qubits and superconducting circuits. Let's start by substituting this relation in the Lagrangian of the Josephson function and its corresponding equation of motion.

$$\mathcal{L} = \frac{1}{2}C\left(\dot{\Phi}\right)^2 + E_{J0}\cos\left(\frac{2\pi\Phi}{\Phi_0}\right) \quad (25)$$

Lagrangian of the system, where. $\frac{1}{2}C\dot{\phi}^2$is the kinetic term, representing the charging energy, and $E_{J0}\cos(2\pi\Phi/\Phi_0)$ is the potential term, representing the Josephson energy.

Where

$$E_{J0} = \frac{I_c\Phi}{\Phi_0} \quad (26)$$

Josephson Energy after substituting the phase difference. Reminding that I_c is the critical current that allows superconducting behaviour.

The Lagrangian equation of motion therefore takes the form:

$$C\ddot{\Phi} + E_{J0}\frac{2\pi}{\Phi_0}\sin\left(\frac{2\pi\Phi}{\Phi_0}\right) = 0 \quad (27)$$

Adapted form of the Lagrangian of the Josephson junction that shows dependence on magnetic flux.

Hamiltonian of the Josephson Junction

Moving forward, Hamiltonian formalism will be preferred. Let us start by writing the conjugate momentum of flux, p_Φ.

$$p_\Phi = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi} \quad (28)$$

The Hamiltonian can be obtained from the Lagrangian using Legendre transformation.

$$H = p_\Phi \dot{\Phi} - \mathcal{L}$$

$$H = \frac{p_\Phi^2}{2C} - E_{J0}\cos\left(\frac{2\pi\Phi}{\Phi_0}\right) \quad (29)$$

Classical Hamiltonian of the system, obtained by performing the Legendre transformation on the Lagrangian.

Transmon Qubit Hamiltonian

To quantise the system, we must ensure that the charge Q and phase ϕ operators commute. They must satisfy the condition

$$[\phi, Q] = i\hbar \quad (30)$$

Commuting relation between charge and flux.

Before writing out the Hamiltonian final expression, let us define **charging energy**, E_C.

$$E_C = \frac{e^2}{2C} \quad (31)$$

The Hamiltonian can therefore be written as

$$H = 4E_C\hat{n} - E_{J0}\cos\hat{\phi} \quad (32)$$

Hamiltonian of the Transmon Qubit. Since charge depends on the number of Cooper-pairs, n, across the junction, as $Q = 2ne$, it is easier to use the number operator $\hat{n}$.

Transmon Regime

Transmon qubits operate in a regime where the Josephson energy significantly exceeds the charging energy, such that $\frac{E_J}{E_C} \gg 1$. This large ratio makes transmon qubits more resilient to noise from charge fluctuations.

In this limit, the potential term $E_J\cos(\phi)$ becomes neglectable, for phase fluctuations are small. Expanding the cosine term around the point $\Delta\phi = 0$, the Hamiltonian has the following approximation:

$$H \approx 4\,E_C \hat{n}^2 + \frac{1}{2} E_J \hat{\phi}^2 \quad (33)$$

This approximation leads the Hamiltonian to resemble a harmonic oscillator with small anharmonicity, that is, a system which behaviour has a small deviation from perfect harmonic behaviour.

Anharmonicity means energy levels aren't spaced in uniform intervals, where higher energy states typically have reduced spacing between them. This also allows for the system to favour transitions between lower energy levels, or qubit states, in this context.

The energy in this system has the following graphic:

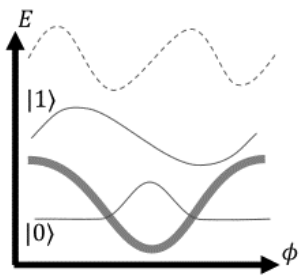

Figure 11 – Energy in a transmon qubit. The darker line corresponds to potential energy.

State Implementation

The energy levels of the approximated Hamiltonian of the system are defined as

$$E_n \approx \left[\sqrt{8\,E_J\,E_C}\left(n+\frac{1}{2}\right) - \frac{E_C}{12}\left(n+\frac{1}{2}\right)^2\right] \quad (34)$$

Energy levels in Transmon Qubits. The first term $\sqrt{8\,E_J\,E_C}$ has behaviour, while the second term introduces small anharmonicity.

Since this anharmonic behaviour, although small, is responsible for making transitions into higher energy levels less likely, it allows the first two energy levels, $|0\rangle$ and $|1\rangle$, to be isolated in a qubit, which is fundamental in qubit operation. It is worth mentioning that anharmonicity, α, can be quantified. The anharmonicity which separates the transition $|1\rangle \rightarrow |2\rangle$ from $|0\rangle \rightarrow |1\rangle$ is, given by:

$$\alpha \approx -E_C$$

Thus, the transmon qubit is formed by the two lower energy states of this Hamiltonian, $|0\rangle$ and $|1\rangle$.

Physical transition

Transition between states is typically driven by applying a **microwave pulse** with a frequency resonant with the energy difference between the two states. The phenomenon that facilitates this transition is known as **resonant excitation**: process of applying an external electromagnetic field at a frequency that matches the energy difference between two quantum states, allowing efficient transitions between those states.

The qubit then absorbs energy from the microwave pulse, which induces a coherent oscillation between $|0\rangle$ and $|1\rangle$ states, governed by the **Rabi oscillation** process. The pulse is timed such that the qubit transitions fully from $|0\rangle$ to $|1\rangle$.

The process is exactly the same for transitioning from $|1\rangle$ to $|0\rangle$.

Rabi Oscillation

Rabi oscillation is the repeated switching of a quantum system between two energy states due to the influence of an external oscillating field, like the microwave pulses aforementioned, when the frequency matches the energy difference between those states.

The probability of finding a qubit in a certain state at time t depends on the Rabi Frequency, Ω_R.

$$\Omega_R = \frac{\mu \cdot E}{\hbar} \quad (35)$$

Definition of the Rabi Frequency, where μ is the transition dipole moment between the two qubit states and E is the amplitude of the microwave field.

The transition dipole moment can be calculated from

$$\mu = \langle 1|\hat{\mu}|0\rangle \quad (\,|0\rangle \rightarrow |1\rangle\,) \quad (36)$$

Transition dipole moment. The dipole moment is calculated using its operator $\hat{\mu} = q\hat{r}$, where q is charge and $\hat{r}$ is the position operator.

We can then write the probability of a qubit state being in the state $|0\rangle$, $P_0(t)$, or state $|1\rangle$, $P_1(t)$, as:

$$P_0(t) = \cos^2\left(\frac{\Omega_R t}{2}\right) \quad (35) \quad \text{and} \quad P_1(t) = \sin^2\left(\frac{\Omega_R t}{2}\right) \quad (37)$$

Since the two states are equiprobable, it is possible to write the general state, **during a transition**, as:

$$|\psi(t)\rangle = \cos\left(\frac{\Omega_R t}{2}\right)|0\rangle + i\,\sin\left(\frac{\Omega_R t}{2}\right) \quad (38)$$

General state of the qubit in terms of Rabi Frequency during a transition.

Implementing State $|0\rangle$

Initializing the ground state depends on the initial state of the qubit. We can have it as a starting point, or we might have to force the system into transitioning from the first state to ground state. Several approaches will be listed here

SYSTEM COOLING

Superconducting qubits are cooled until reaching temperatures in the scale of millikelvin. At these temperatures, most thermal excitations are supressed. **Boltzmann distribution** indicates that while in thermal equilibrium, the probability of being in state $|1\rangle$ becomes negligible compared to being in state$|0\rangle$.

$$P_n = \frac{e^{-\frac{E_n}{kT}}}{Z} \quad (39)$$

Boltzmann distribution. Where E_n is the energy of state n, k is the Boltzmann constant, T is the absolute temperature and Z is the partition function, $Z = \sum_n e^{-E_n/kT}$.

In the limit $T \rightarrow 0$, $P_0 \approx 1$ and $P_1 \approx 0$.

FORCING TRANSITION

If the qubit is in state $|1\rangle$, we can force the transition to the ground state via microwave pulses, like explained above.

A process called **active reset** is a protocol in some qubit systems that, after operations are performed, the system is measured, and feedback mechanisms force it to transition to state $|0\rangle$.

GATE OPERATIONS

During quantum computations, the X gate operation can be performed to flip the qubit from state $|1\rangle$ to $|0\rangle$. This gate simply forces the transition via microwave pulses.

Flux Qubits

This type of superconducting qubit utilizes the quantization of magnetic flux in a superconducting loop that contains at least one Josephson junction. Quantum information is encoded in two distinct states associated with the **direction of the supercurrent** circulating around the loop. The states are treated as flowing **clockwise,** typically $|0\rangle$, and **counterclockwise**, $|1\rangle$.

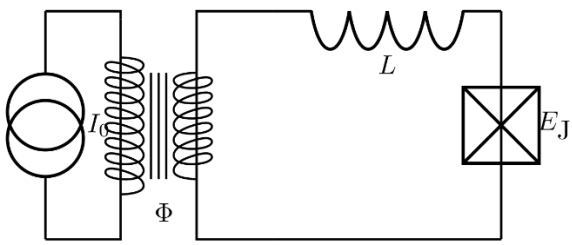

Figure 12 - Circuit diagram representing a flux qubit, where $\boldsymbol{I_0}$ is the superconducting current, $\boldsymbol{\Phi}$ is the magnetic flux threading the loop, $\boldsymbol{E_J}$ represents the energy of the Josephson junction and $\boldsymbol{L}$ is an inductor.

Josephson Junction

Let us use our knowledge of the Josephson junction as starting point.

$$E_J = -E_{J0}\cos(\Delta\phi) \qquad (15)$$

Potential energy in the junction.

$$T = E_C = \frac{Q^2}{2C} \qquad (31)$$

Kinetic energy, which corresponds to the charging energy. It is written with charge Q to acquire a more general form, which will come in handy to further explore charge dependence.

$$\Delta\phi = \frac{2\pi\Phi}{\Phi_0} \qquad (23)$$

Josephson flux-phase relation.

Charge and Magnetic Flux

What makes flux qubits differ from transmon, as both use flux in their expressions? To answer this question, we must take a closer look at the relationship between charge and magnetic flux.

Let us develop equation (23) to get an expression for the **magnetic flux**, Φ. Remember that the magnetic flux quantum, Φ_0, can be defined as

$$\Phi_0 = \frac{h}{2e} \quad (24)$$

Substituting this into equation (23), we get

$$\Delta\phi = \frac{2\pi\Phi}{(h/2e)} = \frac{4\pi e\Phi}{h}$$

The reduced Planck constant is given by

$$\hbar = \frac{h}{2\pi} \quad (40)$$

Meaning

$$h = 2\pi\hbar \quad (41)$$

Using this expression in the development of equation (23), we get

$$\Delta\phi = \frac{4\pi e\Phi}{2\pi\hbar} = \frac{2e}{\hbar}\,\Phi$$

And, therefore

$$\Phi = \frac{\hbar}{2\mathrm{e}}\Delta\phi \quad (42)$$

Relation between the magnetic flux through the loop, Φ, and the phase difference, $\Delta\phi$.

To construct the Hamiltonian, we must understand how the charge Q is related to the magnetic flux Φ through the junction in a flux qubit.

$$Q = -C\frac{d\Phi}{dt} \quad (41)$$

Relation between charge, Q, and magnetic flux, Φ. Notice that the negative sign is merely used as convention.

Simplified Hamiltonian

Since the total Hamiltonian H of the flux qubit can be expressed as the sum of the potential and kinetic energies, we have

$$H = T + U = \frac{Q^2}{2C} - E_J cos\left(\frac{2e}{\hbar}\Phi\right)$$

Substituting the expression (41) for Q, we get

$$H = \frac{1}{2C}\left(-C\frac{d\phi}{dt}\right)^2 - E_J cos\left(\frac{2e}{\hbar}\Phi\right) \quad (42)$$

Similarly to transmon qubits, the charge and phase operators come in handy. Let us define $\hat{Q}$ as the conjugate momentum to $\hat{\phi}$.

$$\hat{Q} = -i\hbar \frac{d}{d\phi} \quad (43)$$

Charge operator, $\hat{Q}$. This expression is better suited when treating the system as a continuous variable system. It also eases calculations in terms of phase.

Remembering the canonical commutation relation $[\phi, Q] = i\hbar$, we may insert expression (43) in the Hamiltonian, deriving it into

$$H = \frac{\hat{Q}^2}{2C} - E_J cos\left(\frac{2e}{\hbar}\Phi\right) \quad (44)$$

The Hamiltonian can be simplified to represent a two-level system which contains the two lowest energy states, $|0\rangle$ and $|1\rangle$. We can do that by using Pauli matrices.

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (45)$$

Pauli x-matrix. Used to represent the tunnelling term, which allows for superposition between the two flux states of the qubit.

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (46)$$

Pauli z-matrix. Used to describe the energy difference between the two classical flux states, distinguishing them.

The Hamiltonian then turns into

$$H = -\frac{E_J}{2}\sigma_z - \frac{E_C}{2}\sigma_x \quad (47)$$

Hamiltonian of the flux qubit, using Pauli matrices. E_J is the Josephson Energy and E_C is the charging energy.

Magnetic Energy Hamiltonian

The previous Hamiltonian expression must be adjusted in a way better considers magnetic energy in the system. We must add the magnetic energy correction term into the potential energy.

The potential energy as a function of magnetic flux is now given by

$$U(\Phi) = -E_J \cos\left(\frac{2\pi\Phi}{\Phi_0}\right) + \frac{(\Phi - \Phi_{ext})^2}{2L} \quad (48)$$

Potential Energy, where Φ is the total magnetic flux in the superconducting circuit, Φ_{ext} is the external magnetic flied applied to the system, L is the inductance of the superconducting loop and, as we've seen before, Φ_0 is the magnetic flux quantum.

It is necessary to introduce two new quantities: the **energy bias**, ϵ, and the **tunnelling amplitude**, Δ. The energy bias in a qubit is what determines the **energetical difference**

between two quantum states. This influences the qubits operation, as it controls the states' relative probabilities during a transition.

$$\epsilon = \frac{E_J}{\Phi_0}\left(\Phi_{ext} - \frac{\Phi_0}{2}\right) \quad (49)$$

Energy bias, ϵ. This expression indicates how the energy difference between the states $|0\rangle$ and $|1\rangle$ changes with the external applied magnetic flux, Φ_0 .

The **tunnelling amplitude** emerges is related to the strength of the quantum **coupling between two flux states**, which correspond to the **two local minima of the potential energy** $U(\Phi)$. It is mainly determined by the height and width of the potential barrier that separates both states and is derived by $U(\Phi)$ when solving, approximately, the Schrödinger Equation for the system.

$$\Delta \sim E_J e^{-\alpha\sqrt{E_J/E_C}} \quad (50)$$

Tunnelling Amplitude, which depends on the Josephson Energy, E_J, charging energy, E_C and α, which is a constant that translates details of potential energy barrier between the states.

Considering these two quantities allows us to get to a final, more precise, Hamiltonian for the flux qubit, which writes as

$$\mathrm{H} = -\frac{\epsilon}{2}\,\sigma_x - \frac{\Delta}{2}\sigma_z \text{ or } \frac{1}{2}(\epsilon\sigma_z + \Delta\sigma_x) \quad (51)$$

Final Hamiltonian of the flux qubit, where ϵ is the energy bias, Δ is the tunnelling amplitude and σ_z and σ_x are Pauli matrices.

The energy in this system has the following graph:

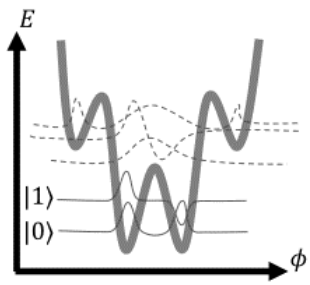

Figure 13 - Energy in a flux qubit. The darker line corresponds to potential energy.

Physical Transition Between States

The two flux states correspond to the two minima of the potential barrier, which can be thought of as the left and right wells in a double-well $U(\Phi)$ potential.

MICROWAVE PULSES

Similar to transmon qubits, it is possible to apply external radiation to force a transition between states. Once again, it must match the energy difference between the states and

EXTERNAL FLUX TUNING

The relative energy difference between the states can also be adjusted by changing the external magnetic flux, since it is proportional to the deviation of the external magnetic flux via equation (49).

$$\epsilon = \frac{E_J}{\Phi_0}\left(\Phi_{ext} - \frac{\Phi_0}{2}\right) \quad (49)$$

At $\Phi_{ext} = \Phi_0/2$, the two states have the same energy: that is, they are degenerate. This allows the qubit to tunnel between $|0\rangle$ and $|1\rangle$ with probability determined by Δ. Adjusting Φ_{ext} therefore controls the relative occupation of each state.

Implementing State $|0\rangle$

Similar to transmon qubits, cooling the system and applying microwave pulses to force transitions are both methods used to implement state $|0\rangle$ is a Flux qubit. Let us look at something specific to this type of qubit.

FLUX CONTROL

Since states are dictated by the flux in the supercurrent, as it controls the energy difference between the states, adjusting the external magnetic flux applied to the qubit helps stabilise it in state $|0\rangle$.

DIVICENZO CRITERIA

Now that we have a basic understanding of how superconducting qubits work, it is possible to understand a system of superconducting qubits satisfy the DiVicenzo Criteria.

Initialising qubits to a simple fiducial state

Like we have seen in the previous chapter, there are many processes available for both transmon and flux qubits to ensure the system is at a specific state.

Long relevant Quantum coherence Times

Both superconducting qubits exhibit coherence times from micro- to milliseconds. This of course varies depending on the materials or design used, but advancements in these sectors allow for qubits to perform a series of quantum algorithms and gate operations without significant degradation from environmental noise or decoherence.

Universal set of Quantum Gates

In transmon qubits, single-qubit gates usually perform by applying microwave pulses at specific **frequencies to rotate the qubit's state** on the Bloch sphere. Two-qubit gates, however, are created by coupling two qubits through a shared resonator or directly via tuneable couplers, enabling entanglement between qubits.

Flux qubits operate similarly, but their control is done by using **magnetic fields to manipulate the supercurrent** in the qubits, and therefore, their states or coupling them, enabling two gate qubit operations.

These single and two-qubit gates, together, are able to assemble a universal gate set which allows any quantum algorithm to be constructed by combining them.

Qubit specific measurement capability

Superconducting qubits have highly controllable non-destructive measurement techniques for reading the state of individual qubits without perturbing others. In transmon qubits, this is done by coupling the qubit to a **microwave resonator**: its state slightly shifts the **resonator's frequency** by probing it with a microwave signal. This shift reveals the state of the qubit without interfering with it, as this change occurs in the resonators.

The flux qubits' state, however, is measured through a device called **SQUID**: Superconducting Quantum Interference Device. This device is sensitive to the **magnetic flux generated by the supercurrent** that circulates in the qubit, which changes depending on its state. Using the SQUID to measure a single qubit's state prevents affecting other qubits in the circuit.

In sum, in both transmon and flux qubits, measurement can be performed in a way that targets single qubits, isolating and coupling them to resonators or SQUIDs, ensuring accurate readouts.

Interconverting stationary and flying qubits and faithfully transmitting flying qubits between specified locations

First of all, what are **flying** and **stationary** qubits? Transmon and flux qubits are categorised as stationary qubits, since they typically **keep their position** in superconducting circuits. Flying qubits, on the other hand, are photons typically within microwave frequency which carry quantum information through transmission lines.

In order to transfer information between the two different types, a stationary qubit is **coupled to a microwave resonator**, which permits the qubit **to transfer its quantum state** to a photon. This process can be **reversed**, so a photon's quantum state is captured by another qubit.

How do superconducting circuits **transmit flying qubits**? Oftentimes the circuits use transmission lines, like waveguides or coaxial cables, that guide the photons between the circuit's nodes. These mechanisms are able to control losses due to signal attenuation or noise, improving fidelity. Other processes like quantum error correction or entanglement distribution ensure this process has high fidelity.

Scalable Physical System with well-characterized qubits

The research on the properties aforementioned has allowed for the precise characterization of the qubits, enabling the construction of large superconducting quantum circuits with clear understanding of each qubit's function in the system.

KEY CONCEPTS

C

Coaxial Cables
Electrical cables consisting of a central conductor, an insulating layer, a metallic shield, and an outer insulating layer, designed to carry high-frequency signals with minimal interference.

Couplers
Devices that allow the strength of interaction between qubits to be adjusted, enabling precise control over entanglement and gate operations in quantum computing.

Crystal Momentum, k

Crystal momentum refers to the effective momentum of an electron moving through a periodic crystal lattice. It is distinct from real momentum but is conserved in the presence of the lattice's symmetry.

Coherent Superposition
A superposition is said to be coherent when the phase between the two or more states is well-defined and stable. This allows for quantum interference effects, such as the formation of distinct, observable interference patterns or the ability to maintain quantum information without losing coherence over time.

E

Energy Bands
Energy bands in a solid represent allowed electron energy levels that result from atoms interacting in a periodic lattice. The distribution of electrons in these bands determines the material's electrical behaviour.

Entanglement Distribution
Creating and sharing quantum entanglement between two or more distant particles or qubits. This enables the particles to remain correlated, meaning the state of one instantly influences the other.

M

Magnetic Flux Quantum, Φ_0
Smallest possible unit of magnetic flux that can exist in a superconductor, equal to approximately $2{,}07 \times 10^{-15}\ Wb$.

P

Pauli Exclusion Principle
Two fermions can't occupy the same quantum state, meaning they must have different sets of quantum numbers. When sharing the same orbital, electrons must have opposite spins, and each state must be occupied by one electron before pairing occurs.

Phonon
The vibrational energy of atomic oscillations in a crystal lattice can be quantized into phonons. A phonon can be thought of as the particle-like unit of sound or heat in a solid material.

Q

Quantum Tunnelling
Quantum phenomenon where particles pass through energy barriers that they classically couldn't, due to their wave-like nature.

R

Resonant
When the frequency of an external signal matches the natural frequency of a system, allowing for maximum energy transfer and efficient interactions between the two.

S

Strong Symmetry
Strong symmetry in a wavefunction

means it remains unchanged or transforms in a predictable way under specific symmetry mathematical operations, (rotations, reflections, or translations, reflecting the underlying symmetry of the physical system). This operations ease solving the system and determining its properties.

U

Unit cell
In crystallography, the unit cell is the simplest portion of the crystal lattice that, when repeated, recreates the entire crystal. It represents the repeating pattern of atoms within the material.

W

Waveguide
Structure that guide electromagnetic waves, like microwave or optical signals, from one point to another with minimal loss. They are typically made of conductive or dielectric materials and are used to efficiently transmit high-frequency signals.

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